Grade 10 | Term 2 | Lesson 1

Outcomes, Support and PMFs

Classroom activity

C1: Characterises discrete random variables in economic and financial contexts by identifying their support, probability mass function and possible outcomes.

Name _____ Date _____

Worked example Notebook sales

A school enterprise offers one notebook to each of three visitors. Record their decisions in order using B for buy and N for no purchase. Each visitor buys with probability 0.4 and decides independently.

An **outcome** records all three decisions. For example, BNN means only the first visitor buys. The random variable X counts buyers, so the distinct outcomes BNN and NBN both give the **value** 1.

The **probability mass function (PMF)** gives $p_X(x) = P(X = x)$. Independence lets us multiply to get $P(\text{BNN}) = 0.4 \cdot 0.6^2 = 0.144$. Adding this mass for each of the three one-buyer outcomes gives $p_X(1) = 0.432$.

Value x (buyers)	**Outcomes giving x**	**Probability calculation**	$p_X(x)$
0	NNN	0.6^3	0.216
1	BNN, NBN, NNB	$3(0.4 \cdot 0.6^2)$	0.432
2	BBN, BNB, NBB	$3(0.4^2 \cdot 0.6)$	0.288
3	BBB	0.4^3	0.064

The **support** is the set of values with positive probability, $\{0, 1, 2, 3\}$. The four masses are non-negative and add to 1. Values outside the support have probability 0.

Guided practice

1. **Payment methods and a fee**

   A shop receives one payment by cash, debit, or credit with respective probabilities 0.25, 0.45, and 0.30. Cash has no fee and either card costs 1 USD. Let F be the processing fee in USD.

   (a) List the payment outcomes and record their probabilities and fees.

Outcome	**Prob.**	**Fee (USD)**
Cash	0.25	0

   (b) State the support. Group the outcomes by fee and show how you find each PMF mass.

f **(USD)**	**Outcomes**	$p_F(f)$ **work**
0		

   Support (USD) _____

   (c) Check that the masses form a valid PMF. Why are there fewer fee values than payment outcomes?

Independent practice

2. **Delivery bonus**

A delivery arrives early, on time, or late with respective probabilities 0.20, 0.50, and 0.30. The bonus B is 2 USD for an early or on-time delivery and -3 USD for a late delivery.

(a) List the possible delivery outcomes. Group them by bonus and state the support of B in USD.

(b) Construct a PMF table for B. Show how you obtained each mass.

(c) Verify the PMF. Explain how a negative bonus can appear in a valid PMF.

3. **Investment gain with a missing mass**

A hypothetical one-month investment has net gain G, in USD. Its only possible values and their masses are shown below.

Net gain g (USD)	-20	0	30
Mass $p_G(g)$	0.20	0.50	q

(a) A student proposes $q = 0.40$. Test the proposal and explain your conclusion.

(b) Find q, check that the completed PMF is valid, and state the support of G in USD.

(c) Find $P(G \geq 0)$. Show which gains you include and explain what the result means.

Solutions

Highlight key Outcome Variable or value Probability Unknown mass

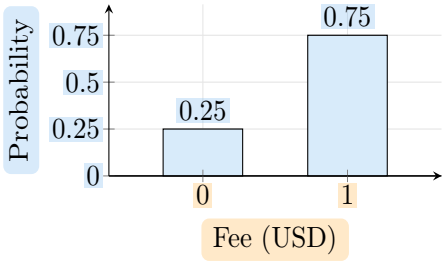
1. **Payment methods and a fee**

   (a) Exactly one payment method occurs. The fee rule assigns the values below.

Payment outcome	Outcome probability	Fee (USD)
Cash	0.25	0
Debit	0.45	1
Credit	0.30	1

   (b) The support of F is $\{0, 1\}$ USD. Add the probabilities of the mutually exclusive card outcomes to get the mass at 1.

f (USD)	Outcomes	$p_F(f)$
0	Cash	0.25
1	Debit, credit	$0.45 + 0.30 = 0.75$



   (c) Both masses are non-negative and $0.25 + 0.75 = 1$. The two card methods are distinct outcomes and both give the same fee, 1 USD.

2. **Delivery bonus**

   (a) The outcomes are early, on time, and late. Early and on time both give 2 USD. Late gives -3 USD. Both bonus values have positive mass, so the support is $\{-3, 2\}$ USD.

   (b) Only a late delivery contributes to $p_B(-3)$. The early and on-time outcomes are mutually exclusive, so their masses add at 2.

Bonus b (USD)	Outcomes giving b	$p_B(b)$
-3	Late	0.30
2	Early, on time	$0.20 + 0.50 = 0.70$

   (c) Both masses are non-negative and $0.30 + 0.70 = 1$. The minus sign in -3 describes the bonus in USD. The chance of that bonus is 0.30.

3. **Investment gain with a missing mass**

   (a) With $q = 0.40$, the total is $0.20 + 0.50 + 0.40 = 1.10$. A PMF must total 1, so the proposal is invalid.

   (b) The mass still needed is $q = 1 - 0.20 - 0.50 = 0.30$. The completed masses 0.20, 0.50, and 0.30 are all positive and sum to 1. The support is $\{-20, 0, 30\}$ USD.

   (c) Non-negative gains include breaking even at 0 USD and gaining 30 USD. Thus $P(G \geq 0) = 0.50 + 0.30 = 0.80$. The investment breaks even or makes a gain with probability 80%.